

directly connected to the processing unit without any additional interfacing electronics, but in case of an analogue sensor, you need additional circuitry to convert the varying voltage levels into a form that the processing unit can use. Also, circuitry is required to amplify the output of the analogue sensor, since this output may not be adequate to be interpreted by the processor. This is basically achieved by using operational amplifiers or op-amps as we have discussed.

Types of Sensors

In robotics projects, the following types of sensors are widely used:

Bump or touch sensors: These are those sensors which are activated when there is touch or bump.

Temperature sensors: These are just like thermometer, and they respond to change in temperature by changing voltage across their terminals. Commonly available temperature sensors are LM35, DS1621 and a thermistor.

Light sensors: These are those sensors which are activated whenever there is significant change in the intensity of light. The most commonly used light sensors are the IR (Infrared) sensors and LDR (Light Dependent Resistor) sensors.

IR Sensors

Appliances such as TVs, music players and ACs have IR transmitter-receiver pairs to transmit signals.

Colour sensors: These are advanced light sensors that can sense even the slightest change in the intensity of light falling on them and thus can detect colours. This is because each colour reflects a different intensity of light when light falls on them. So when there is a change in the intensity of light falling and light reflected, the sensor senses it and sends the signal.

Sonar proximity detector: In this, the reflected sound waves are used to judge distances between the robot and obstacle, with the aid of sound signals.

Speech input or recognition: Various sound/speech patterns or even a human voice can be used to command the robot.

Gas or smoke sensor: These detect noxious or toxic fumes once their concentration in air increases beyond a certain limit.

Pyroelectric infrared sensor: It detects changes in heat patterns over period of time and is thus used as motion detectors.

Sound sensor: Robot responds to sound patterns like a clap or drum beat. The robot can also be tuned to listen to only certain sound frequency (which